Companion XLIII: The Velocity Divergence Field in the Relaxation-Driven Cyclic Cosmology

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Abstract

The velocity divergence field encodes the dynamical response of matter to the gravitational potential and provides a direct probe of the growth of structure. In the standard cosmological model, the divergence evolves according to the linear growth rate and the continuity equation, producing a velocity field whose coherence and amplitude reflect the underlying matter distribution. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the growth of the velocity divergence, the amplitude of bulk flows, and the coupling between density and velocity fields. This Companion develops a continuous description of the RDCC velocity divergence field, deriving how the relaxation scale influences the growth rate, the coherence of large-scale flows, and the dynamical architecture of the cosmic web. The resulting framework provides a new dynamical perspective on RDCC structure formation.

1 Introduction

The cosmic velocity field is a dynamical tracer of the gravitational potential. While the density field describes where matter accumulates, the velocity field describes how matter moves in response to the evolving potential. The divergence of the velocity field, $\theta = \nabla \cdot \mathbf{v}$, plays a central role in structure formation, linking density growth to the temporal evolution of the gravitational field.

In the standard cosmological model, the velocity divergence evolves according to the linear growth rate and the continuity equation. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the growth of the velocity divergence, the amplitude of bulk flows, and the coupling between density and velocity fields. The velocity divergence becomes a direct tracer of the relaxation scale k_{rel}.

2 The RDCC Potential and the Continuity Equation

The continuity equation relates the density contrast δ to the velocity divergence:

$$\frac{\partial \delta}{\partial a} + \frac{1}{aH}\,\theta = 0. \tag{1}$$

The velocity divergence is governed by the Euler equation,

$$\frac{\partial \theta}{\partial a} + \frac{\theta}{a} = -\frac{k^2}{aH}\,\Phi_{\mathrm{RDCC}}. \tag{2}$$

In RDCC, the gravitational potential takes the form

$$\Phi_{\mathrm{RDCC}}(k,a) = \Phi(k,a)\left[1 - \chi_\Phi \left(\frac{k}{k_{\mathrm{rel}}}\right)^\alpha\right]. \tag{3}$$

This modification suppresses the contribution of small-scale modes to the velocity divergence while preserving the coherence of large-scale modes.

3 Growth of the Velocity Divergence Field

The growth of the velocity divergence field is determined by the growth rate,

$$f(a) = \frac{d\ln D}{d\ln a}, \tag{4}$$

where $D(a)$ is the linear growth factor.

In RDCC, the growth rate becomes scale-dependent:

$$f_{\mathrm{RDCC}}(k,a) = f(a)\left[1 - \chi_f \left(\frac{k}{k_{\mathrm{rel}}}\right)^\alpha\right]. \tag{5}$$

Small-scale modes grow more slowly, while large-scale modes retain their coherence. The velocity divergence field becomes smoother and more coherent on scales near k_{rel}, reflecting the temporal structure of the RDCC gravitational field.

4 Bulk Flows and Large-Scale Coherence

Bulk flows arise from the coherent motion of matter on large scales. In RDCC, the enhanced coherence of large-scale modes amplifies bulk flows, while the suppression of small-scale modes reduces small-scale velocity fluctuations.

The bulk flow velocity on scale R is given by

$$v^2_{\mathrm{bulk}}(R) = \int dk\, P_\theta(k)\, W^2(kR), \tag{6}$$

where $P_\theta(k)$ is the velocity divergence power spectrum.

In RDCC, this spectrum becomes

$$P_{\theta,\mathrm{RDCC}}(k) = P_\theta(k)\left[1 - \chi_\theta \left(\frac{k}{k_{\mathrm{rel}}}\right)^\alpha\right]. \tag{7}$$

This modification enhances bulk flows on large scales and suppresses small-scale velocity fluctuations, producing a distinctive RDCC signature.

5 The Dynamical Architecture of the Cosmic Web

The cosmic web is shaped not only by the density field but also by the velocity field. Filaments form where matter flows along coherent velocity streams, while voids expand where the velocity divergence is positive.

In RDCC, the enhanced coherence of large-scale modes produces more ordered flows along filaments and smoother expansion in voids. The velocity divergence field becomes a dynamical map of the relaxation scale, with coherent structures emerging on scales near k_{rel}.

This dynamical architecture provides a new perspective on the formation of the cosmic web in RDCC.

6 Conclusion

The velocity divergence field in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the growth of the velocity divergence, the amplitude of bulk flows, and the coupling between density and velocity fields. These effects produce a distinctive dynamical architecture in the cosmic web, with enhanced coherence on large scales and suppressed fluctuations on small scales. The present Companion establishes the theoretical foundation for future studies of cosmic flows, redshift-space distortions, and the dynamical evolution of large-scale structure in RDCC.

References

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